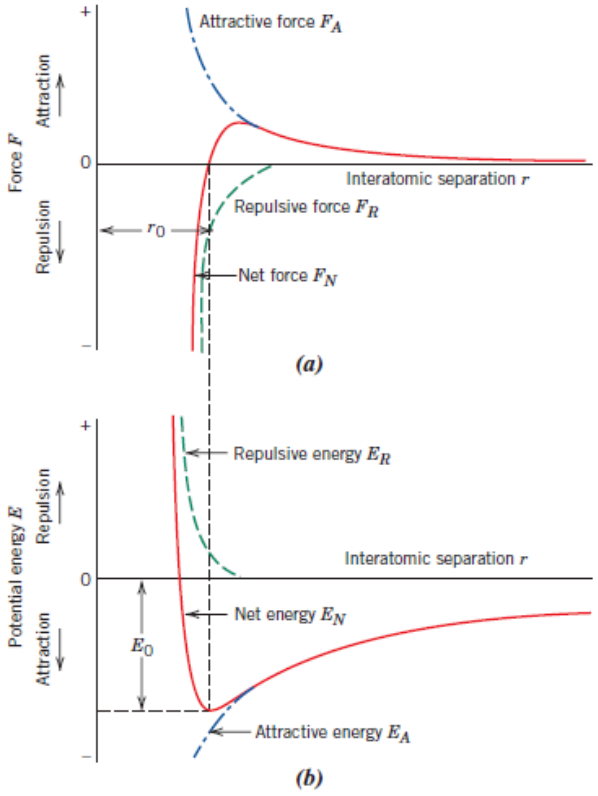
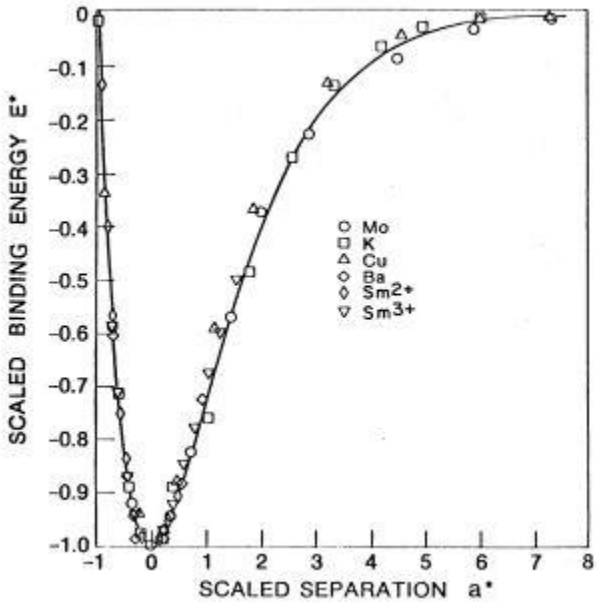
EE2021: Tutorial -1

1. Show mathematically that the existence of molecules, compounds and solids was possible only because the attractive force between atoms (neutral or charged) varies more slowly with interatomic distance than the repulsive force between them





2. Imagine that you are a scholar doing research on a new metal-*Unobtanium*. Your advisor had told you that potential energy vs distance for most metals lie on the same curve if properly scaled/normalized as done above (right). Please come up with a scheme to normalize your experimental data on Unobtanium and verify whether your experimental data is indeed true.

91) $F_{attractive} = \bar{F}_a = -\frac{\lambda_1}{r^a}$

$F_{repulsive} = \bar{F}_b = \frac{\lambda_2}{r^b}$

$$\bar{F} = \bar{F}_a + \bar{F}_R = -\frac{\lambda_1}{r^a} + \frac{\lambda_2}{r^b}$$

$$F(r_0) = 0 \Rightarrow \frac{\lambda_1}{r_0^a} = \frac{\lambda_2}{r_0^b} = k \quad (r_0 = \text{equilibrium seperation})$$

when displaced from r_0, restoring force should pull it back $\Rightarrow \left.\frac{d^2E}{dr^2}\right|_{r=r_0} > 0 \Rightarrow \left.\frac{dF}{dr}\right|_{r=r_0} < 0$

$$\Rightarrow \frac{\lambda_1 a}{r_0^{a+1}} - \frac{\lambda_2 b}{r_0^{b+1}} < 0 \Rightarrow k\left(\frac{a}{r_0} - \frac{b}{r_0}\right) < 0$$

$$\Rightarrow b > a$$

$\Rightarrow$ Exponent of repulsive force > exponent of attractive force

$\Rightarrow$ Repulsive force varies more rapidly than attractive force.

Q2) unobtanium

→ For minimum to be -1 ; $\frac{E_b}{|E_b^{equilibrium}|}$

→ for minima to be at 0; $r - r_e$ for the curve

→ $\frac{E_b}{|E_b^e|} = -1 + \frac{1}{2}\left\{\frac{E_b}{E_b^e|}\right\}''_{r=r_e}(r-r_e)^2 + \cdots$ (Taylor)

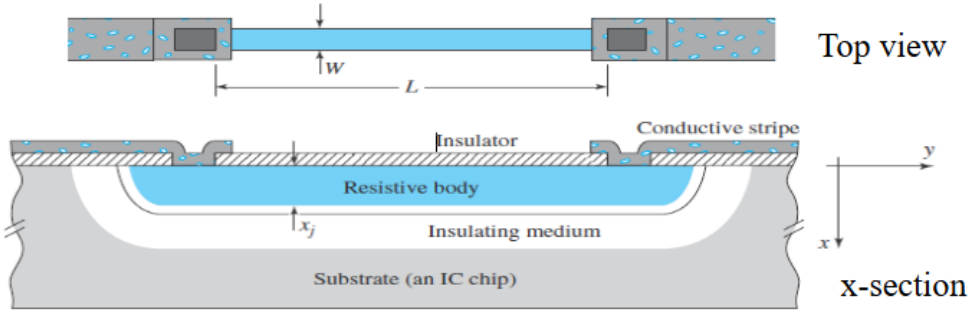
scale it to zero ⇒ so that all of them have same width.

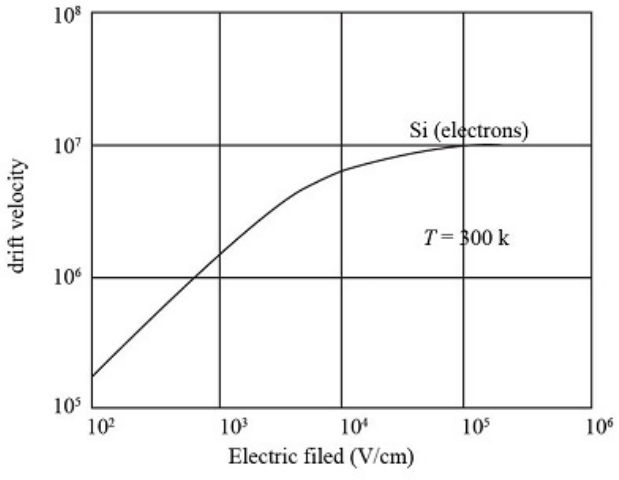
EE2021: Tutorial -2 (August 19th, 2025)

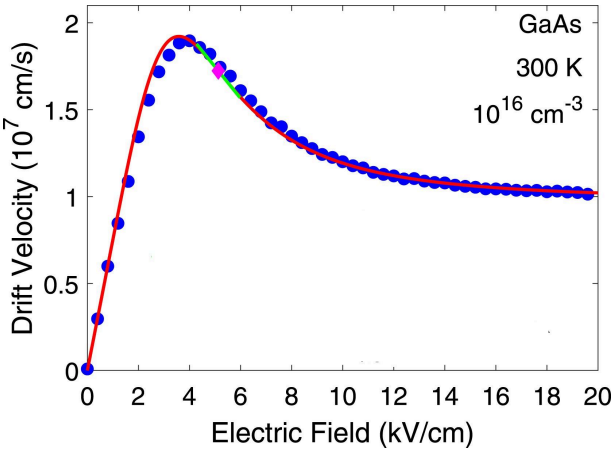
1. One way to make resistors in an IC is to use a doped region of semiconductor as follows. The dimensions are L= 1 µm, x_j=0.1 µm and W=0.5 µm. Two different resistors are made using a) Silicon and b) GaAs. The drift velocity vs electric field plot of electrons in both materials are given. Assume a uniform electron concentration of 10^{16} cm^{-3} in both cases. Plot the I-V characteristics and resistance of both resistors when voltage is varied from 0-15 V. Assume the area of cross-section of the resistor as x_j* W and length as L.

$A = 5 \times 10^{-10}\ cm^2$

$L = 10^{-4}\ cm$







Q1) $J = n e v_d \Rightarrow \frac{I}{x_j * W} = n e v_d$ $\quad E = V/l$

(Si)

$\Rightarrow I = n(x_j * w) e v_d$

low voltage: $v_d \propto E \Rightarrow I \propto V$

high voltage: $(v_d)_{sat} \approx 10^7$ cm/s

$V = 10^5 \times 10^{-4} = 10V$

If voltage > 10V I saturates

(GaAs)

low voltage: directly proportional

peak → saturates

high voltages → current falls

$\Omega(N,q) = \frac{(N+q-1)!}{(q!)(N-1)!} = \frac{(N+q)!}{N!\,q!} \Rightarrow \ln(\Omega) = \ln(N+q)! - \ln N! - \ln q!$

$= (N+q)\ln(N+q) - (N+q) - N\ln N + N - q\ln q + q$

$= (N+q)\ln\left(\frac{N+q}{q}\right) + (N+q)\ln q - N\ln N - q\ln q$

$= (N+q)\left(\frac{N}{q}\right) + N\ln(q/N) = N + \frac{N^2}{q} + N\ln(q/N) \Rightarrow \Omega = e^N \cdot (q/N)^N$

$P! = \sqrt{2\pi P}\left(\frac{P}{e}\right)^P$

$\Rightarrow \ln(P!) \approx P\ln(P/e)$

2. Using Stirling's approximation, derive the expression for the multiplicity of a large Einstein solid as

$$\Omega(N,q) \approx e^{N\ln(q/N)}e^N = \left(\frac{eq}{N}\right)^N \qquad (\text{when } q \gg N)$$

$\bar{E} = \frac{\sum_\gamma E_\gamma e^{-\beta E_\gamma}}{\sum e^{-\beta E_\gamma}} = \frac{1}{Z}\sum_\gamma E_\gamma e^{-\beta E_\gamma} = -\frac{1}{Z}\frac{d}{d\beta}\left(-e^{-\beta E_\gamma}\right) = -\frac{1}{Z}\frac{d}{d\beta}(Z) = -\frac{d}{d\beta}(\ln Z)$

3. Prove that, for any system in equilibrium with a reservoir at temperature T, the average value of the energy is

$$-\frac{\partial}{\partial\beta}\ln Z$$

where β=1/kT

$k = 8.617\times10^{-5}\ eV/K$

4. Imagine a particle that can be in only three states, with energies -0.05 eV, 0, and 0.05 eV. This particle is in equilibrium with a reservoir at 300 K. $\beta = 38.68\ (eV)^{-1}$
   a) Calculate the partition function for this particle.
   b) Calculate the probability for this particle to be in each of the three states.
   c) Calculate the average energy of this particle.
   d) Because the zero point for measuring energies is arbitrary, we could just as well say that the energies of the three states are 0, +0.05 eV, and +0.10 eV, respectively. Repeat parts a)-c) using these numbers. Explain what changes and what doesn't. → partition function & avg energy changes

$Z = \sum e^{-\beta E_\gamma} = e^{0.05(38.68)} + e^0 + e^{-0.05(38.68)} = 8.062$

$P(0.05) = 0.86$
$P(-0.05) = 0.02$
$P(0) = 0.12$

Avg energy $= -0.042$

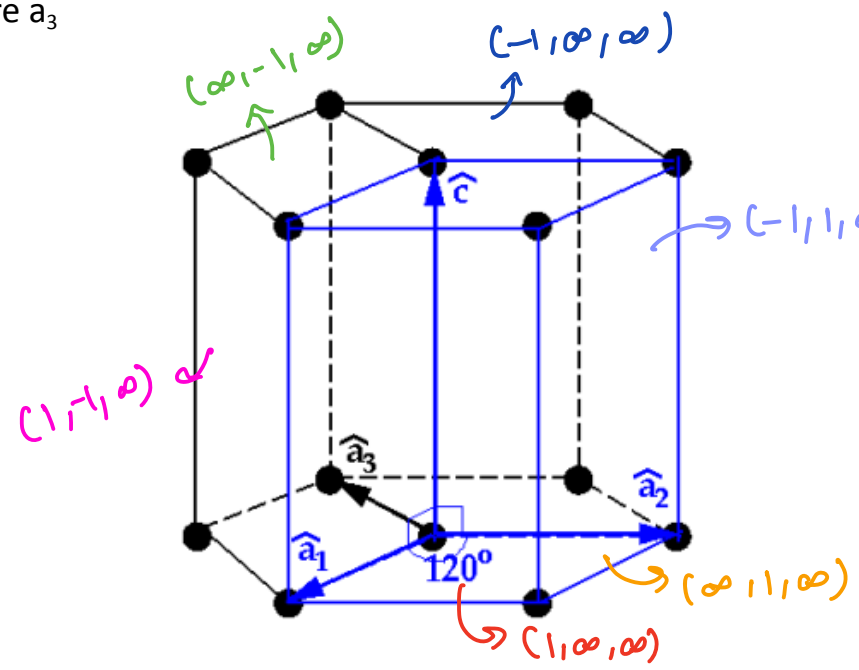
EE2021: Tutorial -3 (August 28th, 2025)

$r = \sqrt{3}a/4 \Rightarrow 2 \times 4/3\pi r^3/a^3 = 8\pi/3 \times 3\sqrt{3}/64 = \pi\sqrt{3}/8$

1. Derive the packing density of BCC and FCC cubic lattice

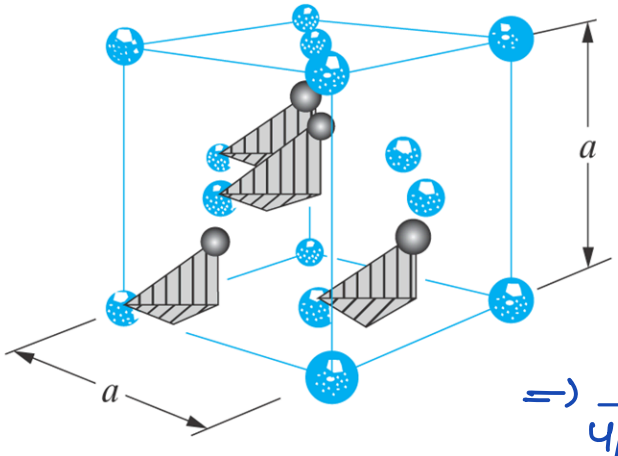
$r = a/2\sqrt{2} \Rightarrow 4 \times 4/3\pi r^3/a^3 = 16\pi/3 \times 1/16\sqrt{2} = \pi/3\sqrt{2}$

2. A hexagonal prism is shown. Find the Miller indices of all six side walls using a_1, a_2 and c as the three coordinate axes as shown below. Ignore a_3



100
010
$\bar{1}$10
$\bar{1}$00
0$\bar{1}$0
1$\bar{1}$0

3. If a=5.43 Å, find the density of atoms in Silicon lattice



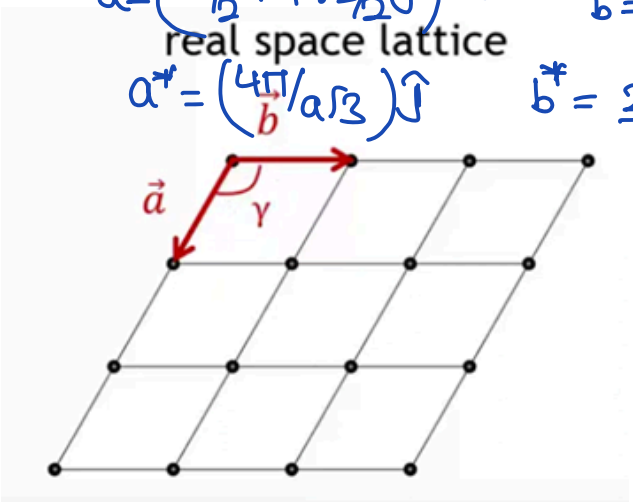
$V = 4/3\pi a^3$

4 internal atoms
+ 8 x 1/8 corners
+ 6 x 1/2 face centered
= 8

$\Rightarrow \frac{8}{4/3\pi a^3} = \frac{6}{\pi a^3}$

4. Write the primitive unit vectors in real space and then find the primitive unit vectors in reciprocal space for this 2-D lattice. Assume $\gamma=120^0$

$a=\left(-\frac{1}{2}\hat{\imath}+\frac{\sqrt{3}}{2}\hat{\jmath}\right)a \qquad b=(\hat{\imath})\,b$



$a^* = \left(\frac{4\pi}{a\sqrt{3}}\right)\hat{\jmath} \qquad b^* = \frac{2\pi}{b}\hat{\imath} + \frac{2\pi}{b\sqrt{3}}\hat{\jmath}$

$a.a^* = 2\pi$

$a.b^* = 0$

$b^*.a = 0$

$b^*.b = 2\pi$

EE2021: Tutorial -4 (Sept 18th, 2025)

1. Consider a system at 300 K which can exchange electrons through an ohmic contact. It consists of three single-particle states, A,B and C whose energy levels are such that $\epsilon_A > \epsilon_B > \epsilon_C$. The probability of A being occupied by a single electron is 0.047. The probability of B and C being empty are 0.5 and 0.047 respectively. Write the grand partition function (GPF) of the system (Quiz1 problem- Please go through Statistical mechanics slides)

2. Which of the following differential equations is linear?

(i) $z\dfrac{d\psi(z)}{dz} + g(z)\psi(z) = 0$, where $g(z)$ is some specific function

(ii) $\psi(z)\dfrac{d\psi(z)}{dz} + \psi(z) = 0$

(iii) $\dfrac{d^2\psi(z)}{dz^2} + b\dfrac{d\psi(z)}{dz} = c\psi(z)$, where b and c are constants

(iv) $\dfrac{d^3\psi(z)}{dz^3} = 1$

(v) $\dfrac{d^2\psi(z)}{dz^2} + \left(1 + |\psi(z)|^2\right)\dfrac{d\psi(z)}{dz} = g\psi(z)$, where g is a constant

3. Consider a one-dimensional particle which moves along the x-axis and whose Hamiltonian is $\hat{H} = -\mathcal{E}d^2/dx^2 + 16\mathcal{E}\hat{X}^2$, where $\mathcal{E}$ is a real constant having the dimensions of energy.

(a) Is $\psi(x) = Ae^{-2x^2}$, where A is a normalization constant that needs to be found, an eigenfunction of $\hat{H}$? If yes, find the energy eigenvalue.

(b) Calculate the probability of finding the particle anywhere along the negative x-axis.

(c) Find the energy eigenvalue corresponding to the wave function $\phi(x) = 2x\psi(x)$.

4. A particle of mass m, which moves freely inside an infinite potential well of length a, has the following initial wave function at $t = 0$:

$$\psi(x, 0) = \frac{A}{\sqrt{a}}\sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{3}{5a}}\sin\left(\frac{3\pi x}{a}\right) + \frac{1}{\sqrt{5a}}\sin\left(\frac{5\pi x}{a}\right),$$

where A is a real constant.

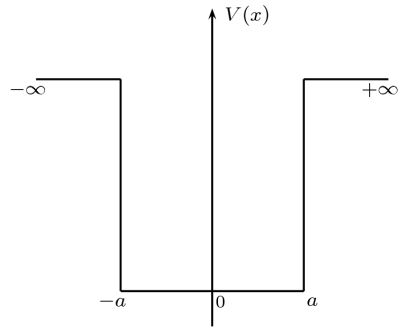
(a) Find A so that $\psi(x, 0)$ is normalized.

(b) If measurements of the energy are carried out, what are the values that will be found and what are the corresponding probabilities? Calculate the average energy.

(c) Find the wave function $\psi(x, t)$ at any later time t.

(d) Determine the probability of finding the system at a time t in the state $\varphi(x, t) = \sqrt{2/a}\sin(5\pi x/a)\exp(-iE_5t/\hbar)$; then determine the probability of finding it in the state $\chi(x, t) = \sqrt{2/a}\sin(2\pi x/a)\exp(-iE_2t/\hbar)$.

5. Consider an infinite potential well of width 2a with a Dirac delta function (Vδ(x)) in the middle of the well. Write the Schrodinger equation, apply boundary conditions and derive the expression for wavefunction and the bound states b) Without solving the Schrodinger equation, try to sketch the lowest two eigenfunctions if the Dirac delta function is replaced by an energy barrier of finite height and width (2b, where b <a)



1. Consider a system at 300 K which can exchange electrons through an ohmic contact. It consists of three single-particle states, A,B and C whose energy levels are such that $\epsilon_A > \epsilon_B > \epsilon_C$. The probability of A being occupied by a single electron is 0.047. The probability of B and C being empty are 0.5 and 0.047 respectively. Write the grand partition function (GPF) of the system (Quiz1 problem- Please go through Statistical mechanics slides)

$f(\epsilon) = \frac{1}{1+e^{(\epsilon-\mu)\beta}}$ $\quad \epsilon_B = \mu$ $\quad f(\epsilon_A) = 0.047$, $f(\epsilon_B) = 0.5$, $f(\epsilon_C) = 0.953$

$0.047 = \frac{1}{1+e^{(\epsilon_A-\epsilon_B)\beta}}$ $\quad \epsilon_A - \epsilon_B = 3KT$

$0.047 = 1 - \frac{1}{1+e^{(\epsilon_C-\epsilon_B)\beta}}$ $\quad \epsilon_B - \epsilon_C = 3KT$

GPF: $\Pi = \sum e^{n(\epsilon-\mu)/KT} = (1+1)(1+e^{3})(1+e^{-3})$

n	$\epsilon - \mu$	
0	0	ϵ_B
1	0	
0	3KT	ϵ_A
1	3KT	
0	-3KT	ϵ_C
1	-3KT	

3. Consider a one-dimensional particle which moves along the x-axis and whose Hamiltonian is $\hat{H} = -\mathcal{E}d^2/dx^2 + 16\mathcal{E}\hat{X}^2$, where $\mathcal{E}$ is a real constant having the dimensions of energy.

(a) Is $\psi(x) = Ae^{-2x^2}$, where A is a normalization constant that needs to be found, an eigenfunction of $\hat{H}$? If yes, find the energy eigenvalue.

(b) Calculate the probability of finding the particle anywhere along the negative x-axis.

(c) Find the energy eigenvalue corresponding to the wave function $\phi(x) = 2x\psi(x)$.

(a) $\hat{H} = -\epsilon d^2/dx^2 + 16\epsilon\hat{x}^2$ $\quad \psi(x) = Ae^{-2x^2} \Rightarrow A = \sqrt{2/\sqrt{\pi}}$

$\hat{H}\psi = -\epsilon \frac{d^2\psi(x)}{dx^2} + 16\epsilon x^2\psi(x)$

$= -\epsilon \cdot Ae^{-2x^2}\{-4+16x^2\} + 16\epsilon x^2 \cdot Ae^{-2x^2}$

$= 4\epsilon Ae^{-2x^2} = 4\epsilon \cdot \psi(x)$

$\Rightarrow$ Eigenvalue $= 4\epsilon$

(b) Symmetric about y-axis $\Rightarrow 1/2$

(c) $$\frac{-\epsilon \frac{d^2}{dx^2}\left(2xAe^{-2x^2}\right) + 16\epsilon x^2\{Ae^{-2x^2}\}\cdot 2x}{}$$

$$= -2A\epsilon \cdot e^{-2x^2}\{-8x - 4x + 16x^3\} + 32\epsilon x^3\{Ae^{-2x^2}\}$$

$$= 24xA\epsilon e^{-2x^2} = 12\epsilon \cdot \psi$$

$\Rightarrow$ energy eigenvalue $= 12\epsilon$

4. A particle of mass m, which moves freely inside an infinite potential well of length a, has the following initial wave function at $t = 0$:

$$\psi(x, 0) = \frac{A}{\sqrt{a}} \sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{3}{5a}} \sin\left(\frac{3\pi x}{a}\right) + \frac{1}{\sqrt{5a}} \sin\left(\frac{5\pi x}{a}\right),$$

where A is a real constant.

(a) Find A so that $\psi(x, 0)$ is normalized.

(b) If measurements of the energy are carried out, what are the values that will be found and what are the corresponding probabilities? Calculate the average energy.

(c) Find the wave function $\psi(x, t)$ at any later time t.

(d) Determine the probability of finding the system at a time t in the state $\varphi(x, t) = \sqrt{2/a} \sin(5\pi x/a) \exp(-iE_5 t/\hbar)$; then determine the probability of finding it in the state $\chi(x, t) = \sqrt{2/a} \sin(2\pi x/a) \exp(-iE_2 t/\hbar)$.

(a) $$\int_0^a \psi(x,0)\,\psi^*(x,0)\,dx = 1 \Rightarrow \frac{A^2}{a}\cdot\frac{a}{2} + \frac{3}{5a}\cdot\frac{a}{2} + \frac{1}{5a}\cdot\frac{a}{2} = 1$$

$$\Rightarrow A^2/2 + 2/5 = 1 \Rightarrow A^2 = 6/5$$

(b) $E_1 = \frac{\pi^2\hbar^2}{2ma^2}$ $\quad E_3 = \frac{9\pi^2\hbar^2}{2ma^2}$ $\quad E_5 = \frac{25\pi^2\hbar^2}{2ma^2}$

$P(E_i) = |c_i|^2$